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Semester: III	**Year : AY 2026-27**
Subject: Programming with Java	
ASSIGNMENT No :	
TITLE : JDBC DriverManager, Connection, and Statement	

Problem Statement

Demonstration and exploration of JDBC classes and interfaces (DriverManager, Connection, Statement).

Learning Objectives

To understand and use the core JDBC classes for database connectivity.

Programme

```
import java.sql.Connection;
import java.sql.DriverManager;
import java.sql.Statement;
import java.sql.SQLException;

public class JDBCExample {

   public static void main(String[] args) {

      String url = "jdbc:mysql://localhost:3306/college";
      String user = "root";
      String password = "Husam123";

      try {
         // Establish database connection
         Connection con = DriverManager.getConnection(
            url, user, password
         );

         System.out.println("Database connected successfully.");

         // Create Statement object
         Statement stmt = con.createStatement();

         // Execute SQL query
         String query = "CREATE TABLE IF NOT EXISTS student (" +
```

```
                "id INT PRIMARY KEY, " +
                "name VARCHAR(50), " +
                "marks DOUBLE)";

        stmt.executeUpdate(query);

        System.out.println("Table created successfully.");

        // Insert a record
        String insertQuery =
           "INSERT INTO student (id, name, marks) " +
           "VALUES (101, 'Husam', 85)";

        int result = stmt.executeUpdate(insertQuery);

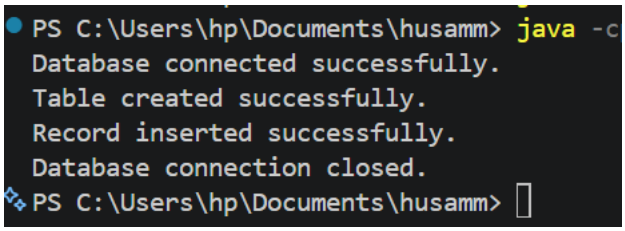
        if (result > 0) {
           System.out.println("Record inserted successfully.");
        }

        // Close resources
        stmt.close();
        con.close();

        System.out.println("Database connection closed.");

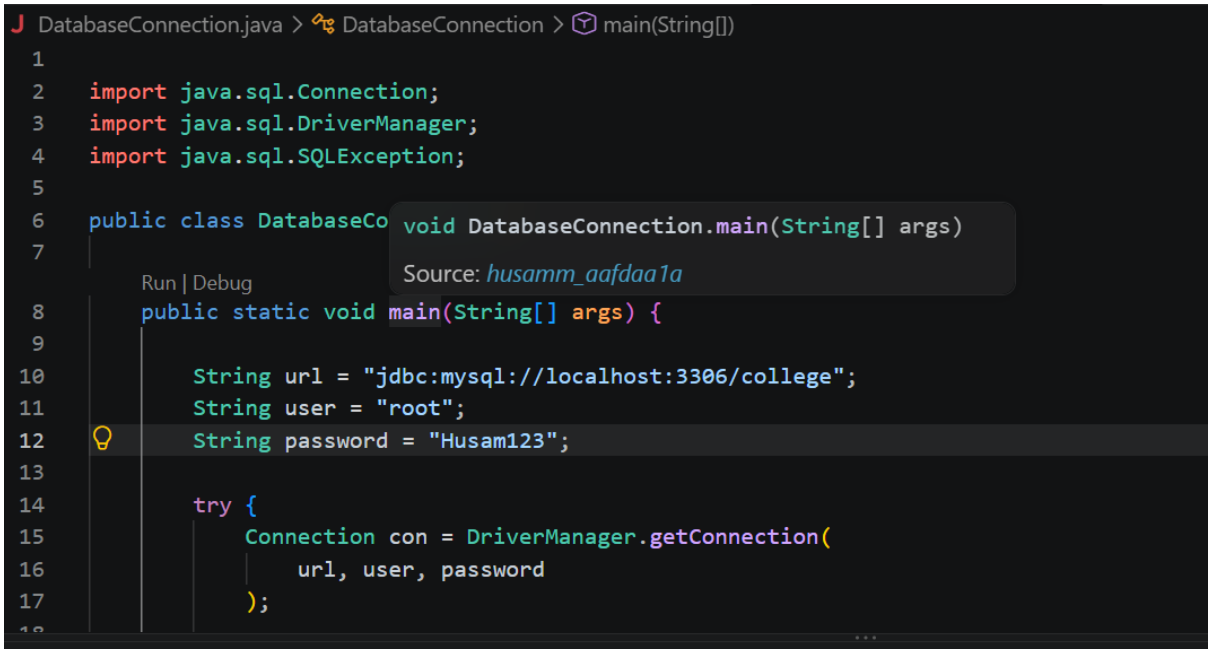
     } catch (SQLException e) {
        System.out.println("Error: " + e.getMessage());
     }
  }
}
```

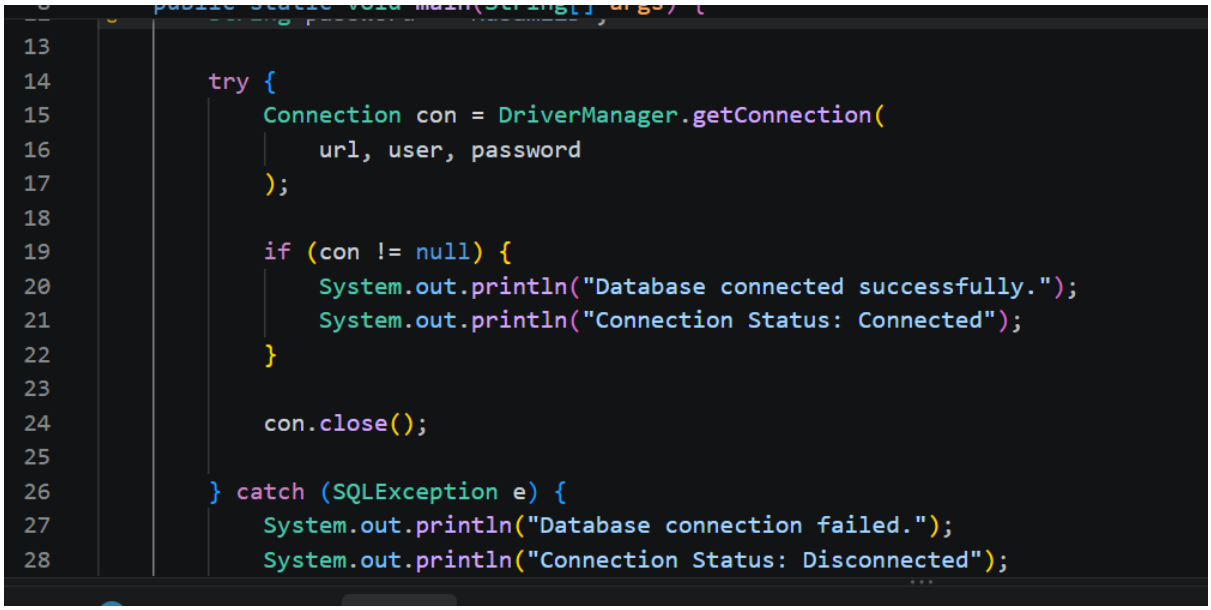
Output



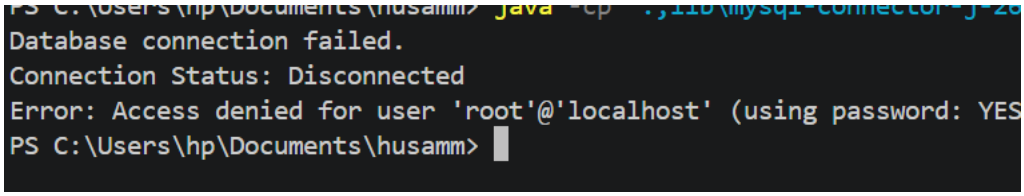
Exercise 1. Create a program to connect Java with database and display database connection status.

Code





Output



2. Create a Java program to connect with a database and display the connection status. After successful connection, show a message indicating that the Student database is connected successfully.

Code

```java
import java.sql.Connection;
import java.sql.DriverManager;
import java.sql.SQLException;

public class StudentDatabase {

    Run | Debug
    public static void main(String[] args) {

        String url = "jdbc:mysql://localhost:3306/college";
        String user = "root";
        String password = "Husam123";

        try {
            Connection con = DriverManager.getConnection(
                url, user, password
            );

            System.out.println(
                "Student database connected successfully."
            );
            System.out.println("Connection Status: Connected");

            con.close();

        } catch (SQLException e) {
            System.out.println(
                "Student database connection failed."
            );
            System.out.println("Connection Status: Disconnected");
            System.out.println("Error: " + e.getMessage());
        }
    }
```

Output

```
PS C:\Users\hp\Documents\husamm> java -cp ".
Student database connected successfully.
Connection Status: Connected
PS C:\Users\hp\Documents\husamm>
```

Conclusion

The program demonstrates database connectivity using JDBC. After successfully connecting to MySQL, it displays a message confirming that the Student database is connected successfully.